A Machile Learning Model for Enthalpies of Formation

J. Cesar Cruz

July 2026

Abstract

This work investigates the prediction of $\Delta_f H^\circ(298.15\,\mathrm{K})$ using molecular descriptors and machine-learning models trained on data from the Active Thermochemical Tables. The workflow included data cleaning, removal of atomic and highly charged ionic species, retrieval of PubChem identifiers and SMILES, and construction of atom-count, bond-count, Morgan-fingerprint, RDKit, and ChemML/Mordred representations. Linear regression models achieved $R^2 \approx 0.71$, whereas nonlinear tree-based models exceeded $R^2 = 0.90$ on a fixed 80/20 random split. The best-performing model was XGBoost, with an RMSE of 155.38 kJ mol^{-1}, an MAE of 95.75 kJ mol^{-1}, and $R^2 = 0.92165$. Feature analysis identified molecular charge as an important predictor; removing it increased the Random Forest RMSE from 172.57 to 333.35 kJ mol^{-1}. These results show that nonlinear models can capture meaningful relationships between molecular structure and thermochemical stability. However, the remaining errors and possible leakage of related species across a random split indicate that grouped validation and uncertainty-aware modeling are necessary to assess chemical transferability.

1 Introduction

Standard enthalpies of formation, $\Delta_f H^\circ$, connect molecular structure and composition with macroscopic thermochemistry. They are required to calculate reaction and combustion energetics, equilibrium predictions, kinetic mechanisms, and the validation of electronic-structure methods. However, reliable reference values are difficult to obtain, particularly for unstable, ionic, or short-lived elements, because they often depend on combining experimental measurements with theoretical calculations. Making a careful evaluation and reconciliation of multiple thermochemical sources is necessary.[1].

The Active Thermochemical Tables (ATcT) address this problem by statistically analyzing and solving a thermochemical network rather than evaluating each species independently. This approach produces formation enthalpies that are accurate, internally consistent, and traceable to multiple experimental and theoretical determinations [2]. The chemical diversity of ATcT makes it a valuable dataset for machine learning because it includes stable molecules, radicals, ions, and transient species over a broad thermochemical range. At the same time, this diversity introduces challenges associated with different charge states, molecular sizes, bonding environments, structural ambiguity, and extreme target values.

Machine-learning methods provide a potential approach for estimating formation enthalpies when experimental measurements or high-level calculations are unavailable. Molecular structures can be represented using interpretable composition and bond-count features, circular Morgan fingerprints [3], or larger collections of molecular descriptors such as those calculated with Mordred [4]. Linear models test whether formation enthalpies can be approximated as additive combinations of molecular features, whereas tree-based methods such as Random Forest [5] and XGBoost [6] can describe nonlinear relationships among composition, bonding, charge, and thermochemical stability.

Nevertheless, strong performance on a random train–test split does not necessarily demonstrate transferability to new chemical families. Structurally related compounds or different charge states of similar species may occur in both subsets, leading to overly optimistic performance estimates. Evaluation must therefore consider absolute prediction errors, neutral and ionic subsets, residual distributions, feature importance, and the chemical similarity between training and test species.

The objective of this project is to develop a transparent and reproducible workflow for predicting $\Delta_f H^\circ(298.15\,\mathrm{K})$ for ATcT molecular species. Atom counts, bond counts, molecular properties, Morgan fingerprints, RDKit descriptors, and ChemML/Mordred descriptors[7] are compared using linear and nonlinear regression models. The study quantifies out-of-sample errors, examines performance for neutral and ionic species, evaluates the importance of molecular charge, and identifies descriptors associated with model generalization. The resulting baseline-to-ensemble comparison is intended to assess both predictive accuracy and the limitations of machine-learning models applied to chemically heterogeneous thermochemical data.

2 Methodology

2.1 Data preparation and curation

Thermochemical data were obtained from an export of the Active Thermochemical Tables (ATcT) [2]. The original spreadsheet contained records distributed across multiple physical rows. The contents of these rows were combined to reconstruct one record per chemical species.

The reconstructed dataset contained 3,442 entries. The standard enthalpies of formation at 0 and 298.15 K were converted to numerical values and expressed in kJ mol^{-1}. No missing values were found for the target quantity, $\Delta_f H^\circ(298.15\,\mathrm{K})$.

Atomic and molecular species were separated based on the number of atoms identified in each chemical formula. Element symbols were validated against the periodic table to prevent state or symmetry labels from being interpreted as chemical elements. Species containing more than one atom were assigned to the molecular dataset, whereas isolated atoms and atomic ions were stored separately. The resulting datasets contained 3,268 molecular species and 174 atoms or atomic ions. Molecular ions and radicals were retained because they form an important part of the chemical diversity of ATcT.

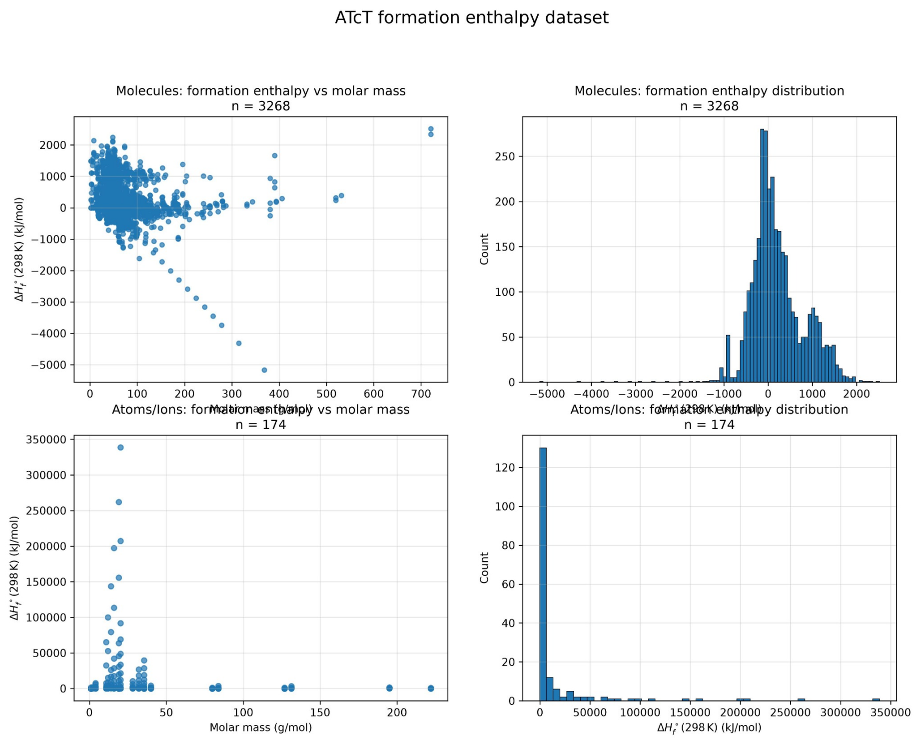

Figure 1: Distribution of the standard enthalpies of formation in the curated ATcT dataset. The left panels show $\Delta_\mathrm{f} H^\circ(298.15\,\mathrm{K})$ as a function of molar mass, while the right panels show the corresponding target-value distributions. Molecular species are presented in the upper panels, whereas atoms and atomic ions are presented in the lower panels.

Figure 1 summarizes the thermochemical distributions obtained after the initial data curation. Molecular species span a broad range of molar masses and formation enthalpies, although most values are concentrated near the center of the distribution. Several molecular entries exhibit substantially negative or positive formation enthalpies and were retained because they represent valid regions of the ATcT thermochemical space. In contrast, the atomic and ionic subset contains a strongly right-skewed distribution with several extremely large positive values. These species were analyzed separately because they do not possess molecular bond representations, and their thermochemical scale differs substantially from that of the molecular dataset.

2.2 Molecular structure assignment

Molecular identifiers and structural representations were retrieved from PubChem using a hierarchical search procedure. Each species was first searched using its CAS registry number, followed by its cleaned chemical name and molecular formula when the previous query was unsuccessful. Phase and electronic-state annotations were removed only from the search strings and remained available in the original ATcT metadata.

For each successful query, the PubChem compound identifier, canonical SMILES, isomeric SMILES, and InChI representation were stored. For unresolved charged species, an additional search using the corresponding formula without the charge annotation was used to identify a possible neutral structural analogue.

Valid structural information was obtained for 3,210 of the 3,268 molecular entries, corresponding to approximately 98.23% of the molecular dataset. The remaining 58 entries primarily included weakly bound complexes, unusual ionic species, hydrates, and species with formulas that did not uniquely define a molecular structure. Entries without a valid SMILES representation were excluded from models requiring bond-based descriptors rather than being represented as molecules with zero bonds.

2.3 Atom-count and bond-count descriptors

Atom-count descriptors were generated directly from the cleaned molecular formulas. For each element A, the corresponding feature n_A was defined as the number of atoms of that element in the species. The complete atom representation was therefore written as

$$\mathbf{x}_{\text{atom}} = (n_{\text{H}}, n_{\text{C}}, n_{\text{N}}, n_{\text{O}}, \ldots) \,. \tag{1}$$

Molar mass and molecular charge were added as separate molecular features. Bond-count descriptors were generated from canonical SMILES using RDKit. Each SMILES string was converted into a molecular graph, and implicit hydrogen atoms were made explicit before counting bonds. Every bond was classified according to the two connected elements and its bond order. The supported bond classes were single, double, triple, and aromatic, represented by the labels `S`, `D`, `T`, and `A`, respectively.

For molecule i, a bond-count feature was defined as

$$n_{AB}^{(t)}(i) = \sum_{b \in \mathcal{B}_i} \mathbb{I}\left[b \text{ connects } A \text{ and } B \text{ with bond type } t\right], \tag{2}$$

where $\mathcal{B}_i$ is the set of bonds in the molecule i, t denotes the bond order, and $\mathbb{I}$ is the indicator function. The bond dictionaries generated for all species were combined into a single feature matrix.

3 Results

3.1 Model performance

The curated bond-feature table contained 3,210 molecular records. An 80/20 random split produced 647 test molecules for the bond/composition models and 643 test molecules for models using the complete descriptor set.

Table 1 compares the principal regression models. The linear baselines gave similar results, with test RMSE values near 301 kJ mol^{-1} and $R^2 \approx 0.71$. This indicates that a purely additive relationship between the descriptors and $\Delta_f H^\circ(298.15\,\mathrm{K})$ was insufficient for the chemical diversity of the dataset. Random Forest reduced the RMSE to 167.57 kJ mol^{-1} using the interpretable bond/composition features. Combining Morgan fingerprints, molecular properties, charge, atom counts, and Mordred descriptors reduced the Random Forest RMSE further to 157.63 kJ mol^{-1}.

Table 1: Test-set performance of the principal regression models. RMSE and MAE are reported in kJ mol^{-1}. The bond/composition and complete descriptor calculations contain slightly different test-set sizes because invalid molecular structures and failed descriptor calculations were removed.

Model	Representation	N_{test}	RMSE	MAE	R^2
OLS	Bond/composition	647	301.77	244.84	0.7096
Ridge	Bond/composition	647	301.20	244.88	0.7107
Random Forest	Bond/composition	647	167.57	99.50	0.9104
Random Forest	All descriptors	643	157.63	91.03	0.9194
Extra Trees	All descriptors	643	157.58	88.34	0.9194
XGBoost	All descriptors	643	**155.38**	95.75	**0.9216**

XGBoost produced the lowest RMSE and highest coefficient of determination, with RMSE = 155.38 kJ mol^{-1}, MAE = 95.75 kJ mol^{-1}, and R^2 = 0.92165. Extra Trees achieved a nearly identical RMSE and the lowest MAE (88.34 kJ mol^{-1}), showing that the identity of the "best" model depends on whether large errors or typical absolute errors are emphasized. Figure 2 shows the substantial improvement of the tree-based models over the linear baselines.

3.2 Influence of molecular charge

Molecular charge was the dominant feature in the Random Forest analysis, accounting for approximately 58.7% of the total impurity-based feature importance. Molar mass and the number of H–O single bonds were the next most important descriptors, with importances of 10.8% and 9.9%, respectively. The

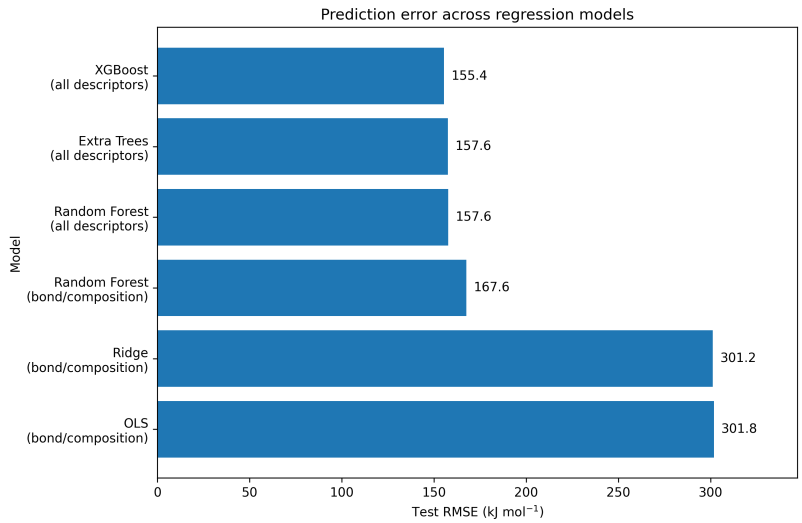

Figure 2: Test RMSE for the principal regression models. Lower values indicate better predictive performance. The nonlinear tree-based models substantially outperform OLS and Ridge regression.

remaining individual bond descriptors contributed much smaller fractions, as shown in Figure 3.

The importance of charge was confirmed through a controlled ablation experiment. Removing charge increased the Random Forest RMSE from 172.57 to 333.35 kJ mol^{-1}, corresponding to an increase of approximately 93%. Table 2 also shows that ionic species had a larger absolute error than neutral species. The ionic subset nevertheless had a higher R^2, which reflects its broader target-value range and should not be interpreted as a lower absolute prediction error.

Table 2: Random Forest sensitivity and subset results. RMSE and MAE are reported in kJ mol^{-1}.

Analysis	RMSE	MAE	R^2
All species, with charge	172.57	101.02	0.9050
All species, without charge	333.35	–	–
Neutral-species model	188.88	90.11	0.7981
Ionic-species model	213.72	136.61	0.8793

Overall, the results demonstrate that nonlinear ensemble models capture relationships that are not represented adequately by an additive linear model. They also show that molecular charge provides essential information for a dataset containing both neutral and ionic species. However, these metrics were obtained from a random holdout split. Structurally related species may therefore occur in both the training and test sets, and the reported values should be interpreted

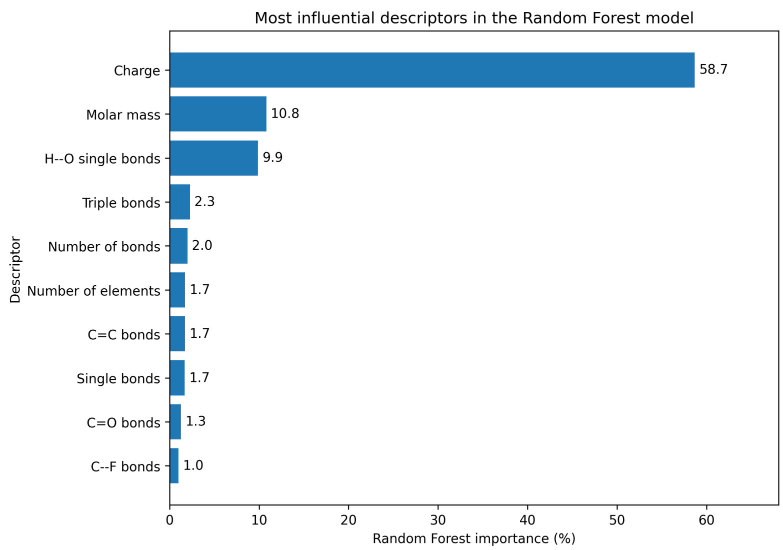

Figure 3: Ten most influential descriptors in the Random Forest analysis. Importance values are normalized impurity-based feature importances. Molecular charge dominates the fitted model.

as random-split performance rather than a definitive measure of transferability to unseen chemical families.

4 Conclusion

This work developed a transparent machine-learning workflow for predicting standard enthalpies of formation, $\Delta_f H^\circ(298.15\,\mathrm{K})$, from molecular composition, bond counts, molecular descriptors, and fingerprints derived from ATcT species. Linear models provided useful baselines but were unable to capture the full chemical diversity of the dataset, yielding $R^2 \approx 0.71$. In contrast, nonlinear tree-based models achieved substantially better performance, with all major ensemble methods exceeding $R^2 = 0.90$.

Among the evaluated models, XGBoost produced the lowest test RMSE, 155.38 kJ mol^{-1}, and the highest coefficient of determination, $R^2 = 0.92165$. Extra Trees achieved the lowest MAE, 88.34 kJ mol^{-1}, indicating that no single metric alone completely determines the preferred model. The results nevertheless show consistently that nonlinear models are more suitable than additive linear approaches for describing the relationships among molecular composition, bonding, structure, charge, and thermochemical stability.

Molecular charge was found to be particularly important. It was the dominant feature in the Random Forest analysis, and removing it increased the RMSE from 172.57 to 333.35 kJ mol^{-1}. This result reflects the broad range of neutral and ionic species represented in ATcT and demonstrates that charge must be treated explicitly when constructing thermochemical machine-learning models.

Future work should therefore include grouped or scaffold-based validation, uncertainty-aware fitting, residual analysis, and systematic testing of descriptor combinations. Despite these limitations, the present study demonstrates that molecular descriptors and ensemble-learning methods can reproduce a substantial fraction of the thermochemical variation in ATcT and provide a useful foundation for more transferable models of molecular formation enthalpies.

References

[1] Branko Ruscic. Uncertainty quantification in thermochemistry, benchmarking electronic structure computations, and active thermochemical tables. *Int. J. Quantum. Chem.*, 114(17-18):1097–1101, 2014.

[2] Branko Ruscic, Reinhardt E. Pinzon, Melita L. Morton, Gregor von Laszevski, Sandra J. Bittner, Sandeep G. Nijsure, Kaizar A. Amin, Michael Minkoff, and Albert F. Wagner. Introduction to active thermochemical tables: Several "key" enthalpies of formation revisited. *J. of Phys. Chem. A*, 108(45):9979–9997, 2004.

[3] David Rogers and Mathew Hahn. Extended-connectivity fingerprints. *J. Chem. Inf. and Model., year = 2010, volume = 50, number = 5, pages = 742–754, doi = 10.1021/ci100050t.*

[4] Hirotomo Moriwaki, Yu-Shi Tian, Norihito Kawashita, and Tatsuya Takagi. Mordred: A molecular descriptor calculator. *J. Cheminformatics*, 10(1):4, 2018.

[5] Leo Breiman. Random forests. *Machine Learning*, 45(1):5–32, 2001.

[6] Tianqi Chen and Carlos Guestrin. XGBoost: A scalable tree boosting system. In *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pages 785–794. Association for Computing Machinery, 2016.

[7] Johannes Hachmann, Mohammad Atif Faiz Afzal, Mojtaba Haghighatlari, and Yudhajit Pal. Building and deploying a cyberinfrastructure for the data-driven design of chemical systems and the exploration of chemical space. *Molecular Simulation*, 44(11):921–929, 2018.